



Forecasting unemployment with internet search data: Does it help to improve predictions when job destruction is skyrocketing?



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ABSTRACT

As more and more daily activities take place online, data on internet behaviour is becoming a key information source. In this sense, several papers have explored the usefulness of internet search data in order to improve the nowcasting and forecasting of economic indicators. Special attention has been paid to predicting unemployment. Nonetheless, most of the empirical evidence on this field has focused in countries with low/moderate rates of unemployment.

This paper follows this line of research and explores the use of Google Trends data in order to improve the forecasting of the figures of unemployment in Spain. This country reveals as a very interesting case due to the sharp increases in unemployment caused by the economic crisis.

With the aim of providing an extensive approach on the Spanish unemployment, we use ARIMA models, also introducing as explanatory variables the Google searches for “job offers” and a business confidence indicator referred to employment perspectives. In this way we combine the time series perspective with qualitative indicators from the supply and the demand sides, leading to a more comprehensive description and also improving forecasting performance.

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1. Introduction

In the last few years the term “big data” has gained popularity in order to refer to the vast amount of digital data which information and communication technologies are making possible to gather, store, diffuse and share in unprecedented ways. Such phenomenon is leading both companies and governments to think carefully over the ways to exploit this data as efficiently as possible.

Hence, economists have been paying special attention to the data coming from online search behaviour since the internet has become a major source of information. Recent figures from Eurostat (2014b) report that 59% of Europeans use the internet to search for information on goods and services; furthermore, among the unemployed Europeans almost half (48%) use the internet to search for a job or to

submit an application, a figure which raises over 80% in countries such as Finland and the United Kingdom.

In this sense, Ettredge et al. (2005, p. 87) early pointed out that “people reveal useful information about their needs, wants, interests, and concerns via their internet behaviour, and that terms submitted to search engines reflect this information”. In fact, search engines are the main entry for internet users when they look for any kind of information as these engines “provide rapid and easy access to websites and information sources” (Eurostat, 2013).

Hence, several papers have addressed the use of the data coming from online search engines in order to improve the nowcasting and forecasting of economic indicators (Choi and Varian, 2009a, 2009b; D'Amuri, 2009; Della Penna and Huang, 2009; Guzman, 2011; McLaren and Shanbhogue, 2011; Vosen and Schmidt, 2011, 2012a, 2012b; Wu and Brynjolfsson, 2010), given that “official economic statistics are generally published with a lag” (McLaren and Shanbhogue, 2011).

Forecasting unemployment with online search-related data has particularly drawn researchers' attention (Anvik and Gjelstad, 2010; Askitas and Zimmermann, 2009; D'Amuri, 2009;

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D'Amuri and Marcucci, 2009; Fondeur and Karamé, 2013; McLaren and Shanbhogue, 2011; Suhoy, 2009). The available empirical evidence shows the usefulness of online search data for nowcasting and forecasting unemployment figures. Nevertheless, it has focused in countries with low/moderate rates of unemployment – France 10.8%, Germany 5.1%, the United Kingdom (UK) 7.2% (Eurostat, 2014a) and the United States (US) 6.7% (Bureau of Labour Statistics, 2014), among others¹.

Within this context, the aim of this paper is to explore whether online search data still helps to improve the nowcasting and forecasting of unemployment in the context of high rates of employment destruction over time. In particular, the present analysis focuses on Spain. This country is an interesting case to study since the economic crisis has been badly hitting the labour market, with rates of unemployment that double the European Union's average. In fact, recent figures released by Eurostat (2014a) show an unemployment rate for Spain of 25.8% compared to the rate of 10.7% estimated for the whole European Union. Only Greece exhibits a higher rate: thus, 27.8% of the Greek economically active population is unemployed.

The analysis presented in this paper is based on a time series approach (ARIMA models), also introducing a business sentiments index (the employment confidence indicator) and the intensity of internet search (through Google Trends). The obtained results confirm the usefulness of internet search data as both an economic indicator and a forecasting tool.

2. Literature review

Over the last few years, several attempts have been made to explore the potential of internet-based search data in order to forecast economic variables. Ettredge et al. (2005) were the first to address such issue and investigated the link between online job searches and the official rates of unemployment in the United States.

Choi and Varian (2009a, 2009b) put forward this line of research by describing and illustrating how internet search data and, in particular, Google's, could be used in order to improve the predictions of several economic indicators such as unemployment claims, retail sales, real estate demand, and holiday destinations' demand.

These two papers have stimulated a lot of recent research in this field. Special attention has been paid to both nowcasting and forecasting unemployment rates with internet job search-related data. Evidence has been provided for different countries: France (Fondeur and Karamé, 2013), Germany (Askatas and Zimmermann, 2009), Israel (Suhoy, 2009), Italy (D'Amuri, 2009), Norway (Anvik and Gjeldstad, 2010), the United Kingdom (McLaren and Shanbhogue, 2011) and the United States (D'Amuri and Marcucci, 2009).

In addition to unemployment figures, the usefulness of internet search-related data has also been explored in other areas such as the housing market. Hence, Chamberlin (2010), McLaren and Shanbhogue (2011) and Wu and Brynjolfsson (2010) have focused on housing prices and sales in the United Kingdom and the United States, while Webb (2009) has examined US home foreclosures.

Private consumption and retail sales have also been of interest to several authors (Carrière-Swallow and Labbé, 2013; Chamberlin, 2010; Della Penna and Huang, 2009; Kholodilin et al., 2010; Jun et al., 2014; Lindberg, 2011; Vosen and Schmidt, 2011, 2012a, 2012b). In this sense, Vosen and Schmidt (2012b) have illustrated how internet search data can be especially useful to predict private consumption in the presence of some irregular events, such as "stimulus" public programmes (i.e. "cash for clunkers" programmes).

Moreover, Guzman (2011) has explored the predictive power of internet queries' data in forecasting inflation expectations, while Artola and Galán (2012) and Bing Pan and Song (2012) have used this data to improve the forecasting performance regarding tourist flows.

Several studies have also investigated whether internet search data might improve the prediction of financial indicators such as investors' attention (Da et al., 2011), stock trade volume and volatility (Joseph et al., 2011; Vlastakis and Markellos, 2012), firms' share prices (Siganos, 2013), foreign currencies (Smith, 2012), and financial market fluctuations (Preis et al., 2010), among others.

Overall, this literature highlights the multiple applications that online search-related data has for forecasting some key economic variables.

3. Data and methodology

3.1. Data

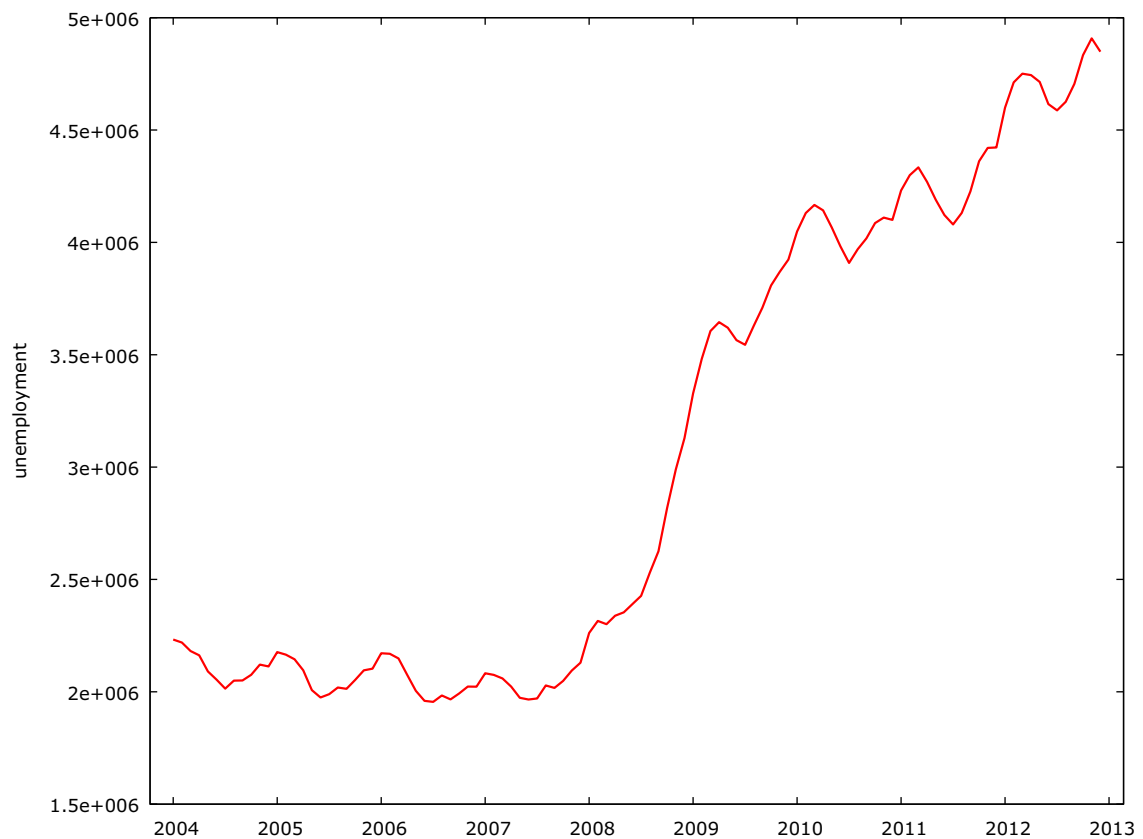
This paper combines variables coming from several data sources. The variable of interest is given by registered unemployment in Spain. This information comes from the Registered Unemployment Statistics, monthly published by the Spanish Ministry of Employment and Social Security (2014), and which provides data on the number of job demands registered by the public employment services. The Spanish labour market can also be monitored through the Economically Active Population Survey (EAPS), which collects quarterly information from a sample of 65,000 households on population's participation in the labour market (National Institute of Statistics, 2004). This paper uses the data from the former database since it allows a larger sample size than the latter (monthly versus quarterly data). The period of analysis is from January 2004 to December 2012, taking the year 2013 as forecasting horizon.

Fig. 1 plots the evolution of registered unemployment in Spain in the period of analysis, January 2004–December 2012. The effect of the economic crisis over the labour market is clearly seen: in mid-2008, unemployment showed a sharp increase that accelerated over the end of that year. While Spain had started the year 2008 with 2,261,925 registered job demands, that figure increased in 1,065,876 along that year, so January 2009 registered a total of 3,128,963 job demands. The increase in job demands over 2007 had only been at 179,417 (January 2008/January 2007).

With the aim of properly analysing (and then forecasting) the evolution of unemployment in Spain, explanatory variables from both the demand and the supply sides of the labour market are taken into account.

On the supply side, data on online searches for job vacancies is incorporated into the model. Following some previous

¹ All these figures refer to December 2013's rates of unemployment.



Source: Ministry of Employment and Social Security (2014)

Fig. 1. Monthly evolution of registered unemployment in Spain (January 2004–December 2012). Source: Ministry of Employment and Social Security (2014).

successful experiences in this field (Choi and Varian, 2009a, 2009b), data from Google Trends' service is used (Google, 2014).² Such service allows to download the data, up to a maximum of five key words, on Google queries. Such data comes as a weekly index that takes value 100 in the week which shows the highest volume of searches for the key words of interest. This data is available from January 2004. For the case of analysis, Spanish unemployment, data was collected on the queries for "oferta de trabajo" and "oferta de empleo", which are two different and common ways to translate "job offer" in Spanish. Since the figures on register unemployment are a monthly time series, data on online searches has been summarized monthly.

As D'Amuri (2009) pointed out, it is necessary to think over some potential limitations of this data. On the one hand, it refers to those who are internet users and employ the Google search engine. However, it could happen that not all unemployed people either use the internet or Google. This does not seem to be the case in Spain given the high figures on internet penetration³ reported by the Spanish National

Institute of Statistics (2014): in 2013, 71.6% of the population between 16 and 74 years old were internet users, a figure that is even higher if only those being part of the economically active population are taken into account⁴; in addition, more than half of the Spanish population was online in 2008 when the effects of the economic crisis started to lead to huge increases in unemployment. Furthermore, the internet has become the main means for job search and even some firms only post their vacancies on the internet and process those curriculums which have been sent online. On the other hand, it should be borne in mind that these online searches might come not only from those who are currently unemployed but also for those who have a job but they would like to change it. In the case of Spain, given the high unemployment rates as well as the predominance of short-term contracts, it is reasonable to think that online searches on job vacancies are mainly carried out by unemployed people and, in a lesser extent, by those who have a job but whose contracts are about to finish. All this suggests that the inclusion of online job-search related data can be a

² It is important to bear in mind that Google leads the market of online search engines, with 88.8% of it belonging to it (Gómez-Uranga et al., 2014).

³ Internet penetration refers to the percentage of population between 16 and 74 years who have use the internet in the last three months.

⁴ The rate of internet penetration for those between 65 and 74 years old is only 22%. Therefore, if this group is not taken into account, the full rate for the Spanish population should be much over the reported 71.6%.

useful tool to explain and forecast the evolution of unemployment in Spain.

On the demand side, that is, regarding employers, the Employment Confidence Indicator (ECI) is included in the analysis. This indicator is published monthly by the Spanish Ministry of Industry, Energy and Tourism (2014) and shows the balance between the positive and negative opinions of industrial firms about the current employment situation and their perspectives three-months ahead. To the authors' knowledge, only D'Amuri (2009) has previously incorporated the demand side of the labour market into the nowcasting of unemployment. In particular, he approaches it by means of the employment expectations index (developed by the European Commission) and the industrial production index. His analysis is carried out for Italy. This country, though being a Southern European country, shows a quite rather different unemployment situation from the one analysed in this paper, Spain, since the Italian unemployment rate is about half of the Spanish.

3.2. Methodology

The analysis of registered unemployment follows the well-known Box–Jenkins approach (Box and Jenkins, 1970), including its identification, estimation and testing. More specifically, the identification of the considered series starts with the stationarity analysis through the augmented Dickey–Fuller unit root test (Dickey and Fuller, 1979) and the KPSS stationarity test (Kwiatkowski et al., 1992). The existence of seasonal unit roots can be checked through the DHF test (Dickey et al., 1984).

Furthermore, the existence of a structural break in the unemployment series should be analysed since it could significantly affect the results of these tests. In fact, in the case of Spanish unemployment the detection of a structural break (dated in March 2008) is particularly important in the identification, estimation and forecasting results.

Once the unemployment series has been transformed in order to achieve the stationarity requirements, and its correlogram has been analysed, the next step is to identify the baseline model, which in our case is given by an ARIMA(0,1,2)(0,1,1) with the following expression:

$$(1-L)(1-L^{12})Y_t = (1-\theta_1 L - \theta_2 L^2)(1-\Theta_1 L^{12})u_t \quad (1)$$

ARIMA models have also been used by Anvik and Gjelstad (2010), D'Amuri (2009), and D'Amuri and Marcucci (2009). All these papers test several models for forecasting the unemployment time series and select ARIMA models since they show the highest performance. In fact, ARIMA models are the most commonly used benchmark models for unemployment forecasting in countries such as the United States (D'Amuri, 2009; Montgomery et al., 1998). The present analysis follows the approach of the aforementioned papers and, after testing several models, the ARIMA(0,1,2)(0,1,1) is selected as the baseline for employment forecasts in the horizon January–December 2013.

Given the jump in Spanish unemployment over the year 2008 (Fig. 1), this first specification (1) can be improved by taking into account the existence of a structural break. In particular, by including a level shift (LS) starting in March 2008

and a level shift with trend (t LS), the baseline B2 expression is derived:

$$(1-L)(1-L^{12})Y_t = (1-\theta_1 L - \theta_2 L^2)(1-\Theta_1 L^{12})u_t + \gamma_1 LS_t + \gamma_2 t LS_t \quad (2)$$

With the aim of providing a more complete description of the evolution of unemployment, explanatory variables can also be included leading to ARIMAX models. More specifically, the considered explanatory variables refer to the previously described ECI and the Google index coming from searches related to job offers with alternative terms.

From the demand side, business' perspectives can be included by adding the ECI as explanatory variable. This leads to the following ARIMAX model M1:

$$(1-L)(1-L^{12})Y_t = (1-\theta_1 L - \theta_2 L^2)(1-\Theta_1 L^{12})u_t + \gamma_1 LS_t + \gamma_2 t LS_t + \beta_1 X_t^{ECI} \quad (3)$$

From the supply side, the perspective of the active population (not only unemployed but also employed people searching for a new job) can be incorporated to our model through the search data from Google Trends. More specifically, data was collected on the queries for “oferta de trabajo” and “oferta de empleo”⁵ leading to models M2 and M3 respectively:

$$(1-L)(1-L^{12})Y_t = (1-\theta_1 L - \theta_2 L^2)(1-\Theta_1 L^{12})u_t + \gamma_2 t LS_t + \beta_1 X_t^{ECI} + \beta_2 X_t^{Google-T} \quad (4)$$

$$(1-L)(1-L^{12})Y_t = (1-\theta_1 L - \theta_2 L^2)(1-\Theta_1 L^{12})u_t + \gamma_2 t LS_t + \beta_1 X_t^{ECI} + \beta_3 X_t^{Google-E} \quad (5)$$

For each of the proposed models, the usefulness of the considered explanatory variables can be evaluated through the regression output and the forecasting results. More specifically, the most commonly used regression indicators have been considered, including the well-known standard deviation of innovations (S) and two informational measures respectively proposed by Akaike (1974) and Schwarz (1978), and given by the following expressions:

$$\text{Standard deviation of innovations : } S = \sqrt{\frac{\hat{u}'\hat{u}}{n-k}} \quad (6)$$

$$\text{Akaike information criterion : } AIC = n \ln(n) + 2k + n(1 + \ln(2\pi)) \quad (7)$$

$$\text{Schwarz information criterion : } SC = -2 \ln L + k \ln(n) \quad (8)$$

Where $\hat{u}_t$ denotes the residuals or innovations, n the sample size and k the number of parameters and L , the Likelihood function.

Regarding the forecasting evaluation, we have considered the most widely used measures of forecasting accuracy including the root mean squared error (RMSE), the mean

⁵ As previously explained in the text, “oferta de trabajo” and “oferta de empleo” are two different and common ways to translate job offer in Spanish.

percentage error (MPE), the mean absolute percentage error (AMPE) and Theil's U index (Theil, 1966). Thus, if we denote by $\hat{Y}_t$ and Y_t the forecasted and the actual values of the considered variable Y for a given period t , the forecasting accuracy can be evaluated through the following expressions:

$$\text{Root mean squared error : RMSE} = \sqrt{\frac{\sum_{t=1}^T (Y_t - \hat{Y}_t)^2}{T}} \quad (9)$$

$$\text{Mean percentage error : MPE} = \sum_{t=1}^T \frac{(Y_t - \hat{Y}_t)}{Y_t T} 100 \quad (10)$$

$$\text{Mean absolute percentage error : MAPE} = \sum_{t=1}^T \frac{|Y_t - \hat{Y}_t|}{Y_t T} 100 \quad (11)$$

$$\text{Theil's U index : U} = \sqrt{\frac{\sum_t \left(\frac{\hat{Y}_t - Y_{t-1}}{Y_{t-1}} - \frac{Y_t - Y_{t-1}}{Y_{t-1}} \right)^2}{\sum_t \left(\frac{Y_t - Y_{t-1}}{Y_{t-1}} \right)^2}} \quad (12)$$

All the proposed measures decrease as forecasts become more accurate, leading to null results when actual and forecasted values are identical.

The expression proposed by Theil (12) provides some additional advantages since its squared expression can be obtained as the addition of three components, respectively related to the bias, repression and disturbance proportion. Furthermore, although this measure is not upper bounded, the case of naïve forecasts ($\hat{Y}_t = Y_{t-1}$) leading to $U = 1$ is usually assumed as a “psychological bound”.

4. Results

The univariate analysis of the monthly unemployment in Spain is significantly affected by the presence of a structural break dated in March 2008. This fact can be clearly seen in the stationarity analysis summarized in Table 1.

More specifically, the results of the augmented version of the Dickey and Fuller unit root test (ADF) and the KPSS stationarity test detect two regular unit roots when the whole sample period 2004:01–2012:12 is considered (second and third columns of Table 1). Nevertheless, these conclusions significantly change when two independent sub-samples (before and after the structural break) are analysed. In this case, as shown in Table 1, the first differenced series of

unemployment $(1-L)Y_t$ can be assumed to be stationary, according to both ADF and KPSS results.

In order to complete the stationarity analysis we have also run the DHF test proposed by Dickey et al. (1984). Since we fail to reject the hypothesis of a seasonal unit root, a seasonal difference has been added to the dependent variable. This leads to the baseline ARIMA(0,1,2)(0,1,1) model B1 (1) and its extension B2 (2), including the structural break through level shift and level shift with trend terms.

The estimation of these two baseline proposals is summarized in Table 2, showing that the introduction of the structural break improves the estimation, leading to lower results for the standard deviation of innovations and the information criteria, and also allowing the assumption of normality for the residuals.

A more comprehensive description of the evolution of Spanish unemployment is provided by the ARIMAX models including explanatory variables from both the demand and the supply sides. With this aim, model M1 (3) includes the ECI computed as the balance between the positive and negative opinions of industrial firms on the current employment situation and perspectives.

As shown in Table 2, a significant negative coefficient is estimated for the ECI, confirming that the more positive the perspectives, the lower the unemployment. Furthermore, since in this case only the interaction term (level shift with trend) related to the structural break results to be significant, the level shift term has been omitted.

As expected, this ARIMAX model leads to a new improvement in the estimation output (the standard deviation of innovations and information criteria decreases with regard to the baseline models) and the estimation residuals follow a Normal distribution.

The last two columns of Table 2 show the corresponding estimation results of the ARIMAX models M2 (4) and M3 (5) which include the supply-side explanatory variables, those being, the Google Trends indicators referred to the queries for “oferta de trabajo” and “oferta de empleo”. These models significantly improve the previous ones; in fact, model M2 results to be the most suitable option according to the significance of the estimated coefficients, the standard deviation of residuals and the information criteria.

With regard to the forecasting results, Table 3 summarizes the main evaluation indicators, including the root mean squared error, the mean percentage errors (computed over the observed forecasting errors and also in absolute terms) and Theil's U index (Theil, 1966) with its three components, respectively related to the bias, repression and disturbance proportions.

According to all these indicators, significant forecasting improvements can be observed in the ARIMAX models when Google Trends variables are included. Such results confirm the usefulness of taking into account internet searches of job offer related terms.

Table 1

ADF and KPSS tests' critical levels for monthly Spanish registered unemployment.

Variable	Sample period 2004:01–2012:12		Sample period 2004:01–2007:12		Sample period 2009:01–2012:12	
	ADF	KPSS	ADF	KPSS	ADF	KPSS
Y_t	$p = 0.68$	$p < 0.01$	$p = 0.25$	$0.01 < p < 0.05$	$p = 0.94$	$p < 0.05$
$(1-L)Y_t$	$p = 0.32$	$0.01 < p < 0.05$	$p = 0.003$	$p > 0.10$	$p = 0.02$	$p > 0.10$

Table 2

Estimation results for ARIMA and ARIMAX models on Spanish unemployment.

	Baseline B1	Baseline B2	Model M1	Model M2	Model M3
θ_1	0.7853 ***	0.7603 ***	0.7422 ***	0.6858 ***	0.6863 ***
θ_2	0.4055 ***	0.4006 ***	0.3888 ***	0.3763 ***	0.3766 ***
Θ_1	−0.4618 ***	−0.5526 ***	−0.5339 ***	−0.6607 ***	−0.6555 ***
γ_1 (Level shift)		58,439.3 **			
γ_2 (Level shift with trend)		−751.266 **	−258.788 **	−339.137 ***	−304.633 ***
β_1 (Employment Confidence Indicator)			−1206.42 ***	−704.939 *	−785.996 *
β_2 (Google index for “oferta de empleo”)				304.563 **	
β_3 (Google index for “oferta de trabajo”)					308.017 *
Standard deviation of innovations	33,237.26	33,043.72	32,428.39	31,212.74	31,598.58
Akaike criterion (AIC)	2259.380	2258.660	2255.088	2249.829	2252.163
Schwarz criterion	2269.595	2273.983	2270.412	2267.706	2270.040
Normality test Chi-square(2)	Chi-2 = 2.57 p = 0.27	Chi-2 = 1.79 p = 0.40	Chi-2 = 1.34 p = 0.51	Chi-2 = 2.41 p = 0.30	Chi-2 = 2.49 p = 0.29

[*** significant at 1% level, ** significant at 5% level, * significant at 10% level]

Fig. 2 plots the obtained predictions for unemployment in the forecasting horizon January–December 2013 with the proposed models.

The comparison of the forecasting performance of our results with similar researches is limited by the lack of previous investigations referred to the same country and forecasting period. Nonetheless, it is worth noting that, in terms of the mean squared error (which is the most widely used indicator), the analyses referred to the United States by D'Amuri and Marcucci (2009) and to Norway by Anvik and Gjelstad (2010), respectively, show that Google-based models improve the forecasting accuracy in 19.8% and 18.3%; while our findings – referred to Spain during a particularly difficult period – provide an improvement of 15% with regard to the baseline model B2 and significantly higher with regard to model B1.

5. Concluding remarks

The last years have witnessed how the internet has become a major element in daily life. Therefore, it has grown to be a major source of information not only because of all the content that is online but also because the activities that take place online can be tracked. Within this context, special attention has started to be paid to the usefulness of the data coming from individuals' internet search behaviour. Since this data reflects people's “needs, wants, interest, and concerns” (Ettredge et al., 2005, p. 87), it holds a great potential to help to improve the nowcasting and forecasting of some key economic-related variables.

Hence, a new line of research has emerged, with its focus on the use of internet-based search data in order to forecast economic variables. This paper has aimed to contribute to this new field of research by paying attention to the use of this data

to the analysis of the labour market, and in particular to the evolution of unemployment in Spain. This country offers a very interesting case of study since it has been badly hit by the economic crisis and, as a consequence, it has been exhibiting one of the highest unemployment levels in the whole European Union since the year 2008. All the previous evidence regarding unemployment has focused on countries with much more stable labour markets.

With this aim, the time series Box–Jenkins approach has been used and complemented with the inclusion of explanatory variables from the demand and the supply sides of the labour market. More specifically, the ECI, published by the Spanish Ministry of Industry, Energy and Tourism, has been included to capture the demand side of the labour market; while the Google Trends indicators related to job search terms (“job offer”) are used as proxy for the supply side. Both variables are found to be statistically significant to explain the evolution of unemployment, and their estimated coefficients show the expected signs: negative in the case of the estimated coefficient for the ECI; and positive for Google search-related indicators.

Hence, the obtained results confirm the potential of the proposed approach, since it provides a comprehensive description of the evolution of unemployment and, furthermore, it significantly improves the estimation and forecasting of unemployment's figures in a context of important economic shocks, such as the ones that the Spanish economy has been experimenting due to the crisis.

Overall, the presented evidence highlights the importance and usefulness of internet search-related data for the nowcasting and forecasting of economic variables. Such results are especially interesting since traditional data collection is costly and many countries face important budgetary restrictions because of the

Table 3

Evaluation statistics of unemployment forecasting in the horizon January–December 2013.

	Baseline B1	Baseline B2	Model M1	Model M2	Model M3
Root mean squared error	219,440	64,065	67,653	61,639	59,056
Mean percentage error	−3.3073	1.2408	0.1319	0.8527	0.6042
Mean absolute percentage error	3.5837	1.2408	1.1794	1.1678	1.075
Theil's U	3.2791	0.9023	0.9707	0.8678	0.8289
Bias proportion	0.5191	0.8899	0.0122	0.4635	0.2593
Regression proportion	0.1412	0.0302	0.0115	0.0340	0.0511
Disturbance proportion	0.3474	0.0798	0.9763	0.5025	0.6896

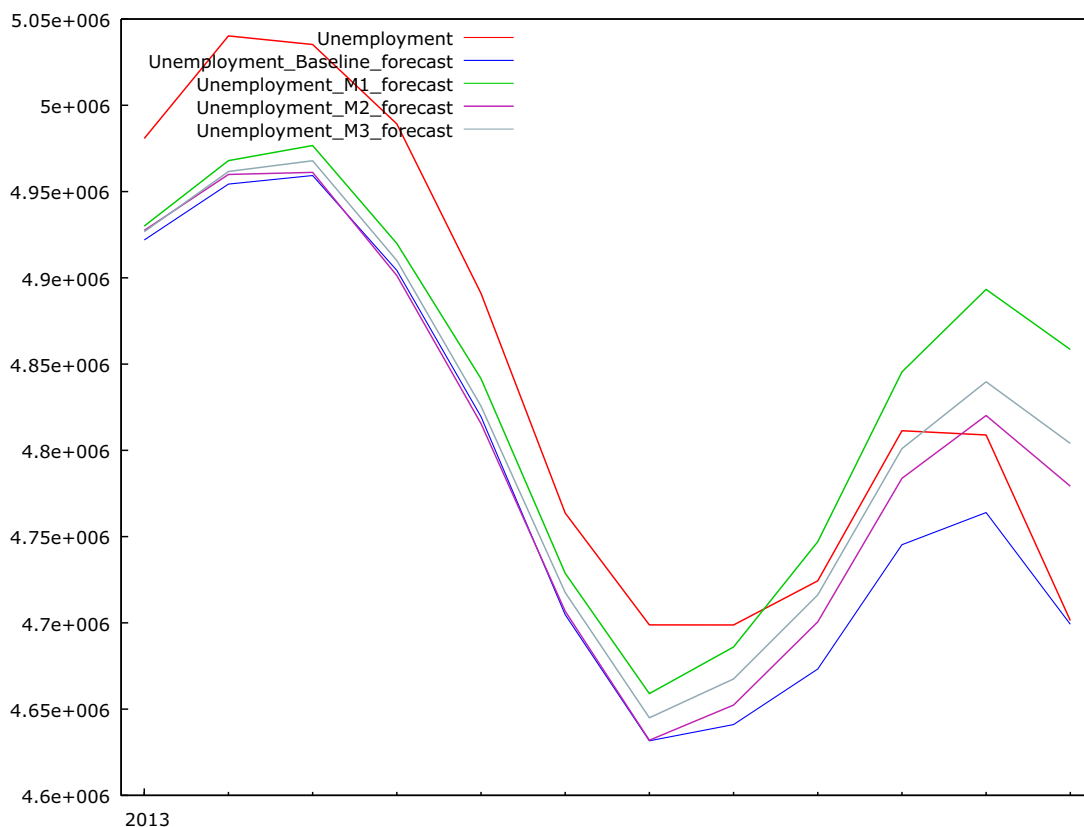


Fig. 2. Actual and forecasted unemployment in the forecasting horizon January–December 2013.

economic crisis that they have not been able yet to overcome. As more and more activities take place online, internet-related data will increase in volume and importance. In this sense, Varian points out (2013, p.27) “since computers are now involved in many economic transactions, big data will only get bigger”. Hence, there is a growing need to further explore the application of these new data sources in order to improve the nowcasting and forecasting of economic statistics.

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